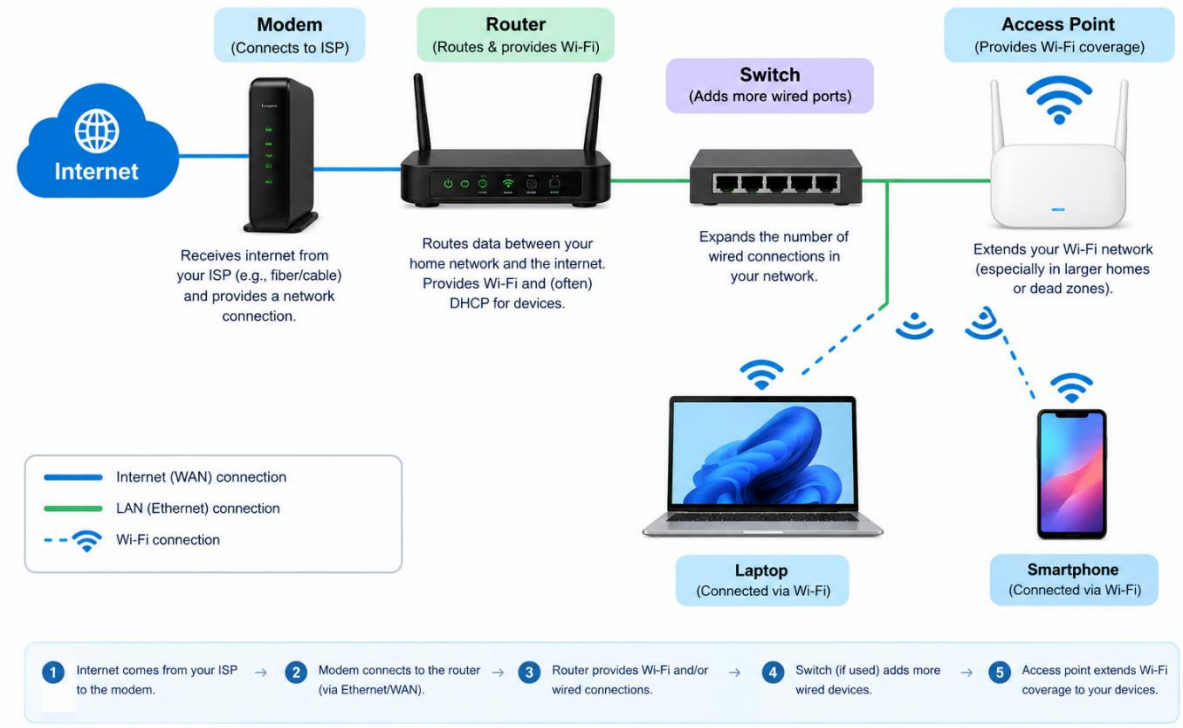
Assignment-4.1

TASK-1

Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



TASK-2

List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Modem (Modulator-Demodulator):

- **Main function**: Converts incoming analog signals from your Internet Service Provider (ISP) into digital data your devices can understand, and vice versa.
- **Real-life example**: Connected to the coaxial cable or phone line coming into your apartment to bring in broadband internet.

Router:

- **Main function**: Connects multiple networks together (such as your home network to the internet) and directs traffic packets to the right destination using IP addresses.
- **Real-life example**: At home to share a single internet connection with your laptop, phone, and smart TV simultaneously.

Switch (Network Switch):

- **Main function**: Connects multiple wired devices together within a Local Area Network (LAN) and uses hardware MAC addresses to send data only to the specific port of the intended recipient.
- **Real-life example**: In a small office or gaming setup to plug in four desktop computers using Ethernet cables so they can talk to each other and share files quickly.

Access Point (Wireless Access Point / WAP):

- **Main function**: Creates a wireless Wi-Fi signal and acts as a bridge that lets wireless devices connect to a wired Ethernet network.
- **Real-life example**: Mounted on the ceiling of a large hotel hallway or university library to give guests wireless internet access from a wired wall drop.

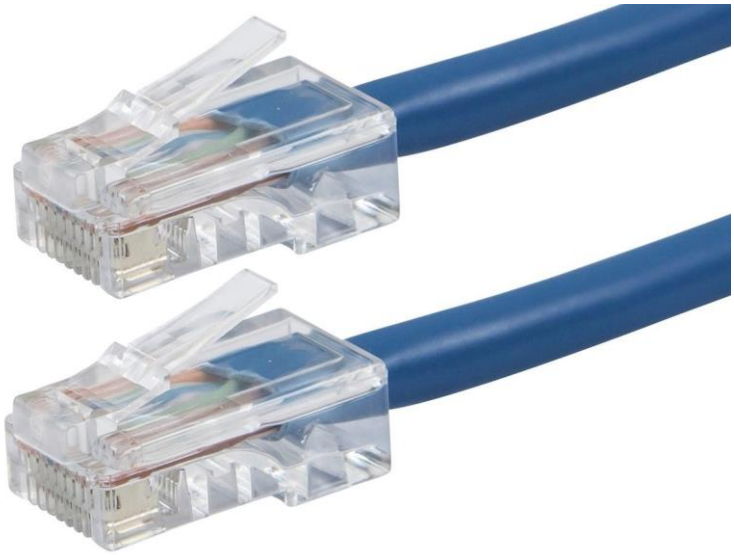
Network Cable:

- **Main function:** Physically carries electrical or optical data signals from one networking device or computer to another with minimal interference.
- **Real-life example:** Running a blue cable from the back of your router straight into your desktop PC or gaming console for a faster, lag-free connection.

TASK-3

Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

1. Ethernet Cable (Twisted-Pair)



Use: Commonly used to connect computers, routers, and switches in home and office networks.

Source: [Cat6 Ethernet patch cable via Amazon]

2. Coaxial Cable



Use: Commonly used for cable television, CCTV systems, and cable broadband internet connections.

Source: RG6 Coaxial Cable via Amazon

3. Fiber-Optic Cable



Use: Commonly used for high-speed internet, telecommunications, and long-distance data transmission.

Source: SC-to-SC Fiber Optic Patch Cable via Amazon

TASK-4

Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Feature	**Router**	**Switch**
Main Use	Connects different networks together (such as a LAN to the WAN/Internet) and directs traffic between them.	Connects multiple end-user devices (computers, printers, servers) within a single local network.
Typical Location	At the edge of a local network, acting as the gateway between the local network and the internet service provider (ISP).	Inside a local network (LAN) to link computers, access points, and devices together, often connected directly to a router.
How They Handle Data	Operates at the Network Layer (OSI Layer 3) and forwards data in the form of packets using IP addresses.	Operates at the Data Link Layer (OSI Layer 2) and forwards data in the form of frames using MAC addresses.
Network & Interface Support	Supports both wired Ethernet connections and wireless (Wi-Fi) signals.	Primarily supports wired Ethernet connections across multiple physical ports (8 to 48+ ports).